



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

CV1010 -1 – SUSTAINABLE WATER AND SANITATION SYSTEM

(For Minor Degree)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II each** and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which of the following is the primary benefit of groundwater recharge?
A) Decreases water table level
B) Improves water quality
C) Increases surface runoff
D) Reduces water infiltration
2. Which technique involves collecting and storing rainwater from surfaces like rooftops?
A) Groundwater pumping
B) Rainwater harvesting
C) Desalination
D) Wastewater reclamation
3. The purpose of a water tariff is primarily:
A) To discourage all water use
B) To recover costs and encourage efficient water use.
C) To fund only irrigation projects.
D) To make water entirely free for all users
4. Roof water harvesting is part of which broader concept?
A) Decentralized wastewater treatment
B) Land treatment systems
C) Water Conservation
D) Low-cost sanitation
5. The 'ability to pay' principle is often considered when determining the structure of a
A) Wastewater treatment plant design
B) Groundwater recharge technique.
C) Water tariff.
D) Rainwater harvesting technology
6. The term 'Water Economics' primarily concerns:
A) The biological quality of water
B) The cost, valuation, and allocation of water resources.
C) The physical movement of water in the hydrological cycle.
D) The chemical treatment of wastewater
7. A low, subsidized water tariff could lead to:
A) Increased water conservation.
B) Overuse and wastage of water.
C) Higher financial sustainability of utilities.
D) Greater investment in new water sources
8. Which factor is least relevant in determining a sustainable water tariff
A) Operation and Maintenance costs.
B) Capital costs of infrastructure.
C) Rainfall intensity in the region.
D) Social equity and affordability.
9. Which parameter is a key indicator of the reliability of a wastewater treatment system?
A) The volume of the reactor.
B) The consistency of effluent quality under varying load.
C) The aesthetic design of the plant.
D) The maximum flow capacity
10. Granular-medium filtration is typically utilized as a stage in which type of treatment?
A) Primary treatment
B) Secondary biological treatment
C) Tertiary treatment
D) Preliminary screening
11. Which of the following is generally considered a highly cost-effective and reliable Natural Wastewater Treatment System?
A) Activated Sludge Process
B) Constructed Wetlands
C) Membrane Bioreactor (MBR)
D) Advanced Oxidation Process (AOP)

12. **Micro-screening is primarily designed to remove:**
 A) Dissolved salts. B) Fine suspended solids.
 C) Heavy metals. D) Pathogenic bacteria
13. **The primary mechanism for pollutant removal in a Constructed Wetland is**
 A) High-pressure physical separation. B) Biological uptake by plants and microbial degradation in the rhizosphere
 C) Chemical precipitation with lime. D) Ultraviolet radiation.
14. **Which design consideration is crucial for the performance of a Natural Wastewater Treatment System (NWTs)?**
 A) High energy input. B) Sufficient land area and hydraulic retention time (HRT).
 C) Use of complex chemical additives. D) High degree of automation
15. **Tertiary treatment processes are mainly focused on removing:**
 A) Large floating debris. B) Settleable organic solids.
 C) Non-biodegradable organics, nutrients, and dissolved inorganics D) Primary sludge
16. **The effectiveness of a sand filter in tertiary treatment depends largely on**
 A) The depth of the filter bed and media size. B) The velocity of wind over the bed.
 C) The of the raw wastewater D) The amount of dissolved oxygen in the influent.
17. **The primary driver for implementing wastewater reuse is typically:**
 A) To meet strict discharge limits into rivers. B) To generate electricity.
 C) water scarcity and demand for non-potable uses. D) To reduce the in the wastewater
18. **An example of a low-cost, dry sanitation method is a**
 A) Septic tank system. B) Aqua privy.
 C) Ventilated Improved Pit (VIP) latrine. D) Conventional sewage system
19. **Which is a key environmental benefit of land treatment systems for wastewater**
 A) High energy consumption. B) Nutrient recycling to benefit vegetation.
 C) High production of chemical sludge. D) Immediate discharge to a surface water body
20. **The technical consideration for wastewater reclamation includes:**
 A) Determining the required effluent quality based on the intended reuse application B) The color of the receiving water body.
 C) The cost of the land for the plant. D) The proximity to a major highway

PART - B: DESCRIPTIVE ANSWER QUESTIONS

	Unit – I	Marks	BT*	CO*	PO*
1. a)	Explain the concept of sustainable water management and detail the importance of Water Conservation in addressing water scarcity challenges	08	L*2	1	1,2
b)	Discuss the economic principles behind the implementation of a Water Tariff. How does a well-designed tariff structure promote financial sustainability?	08	L2	2	1,2,3
2. a)	Describe the technology, quality concerns, and health issues associated with Roof Water Harvesting	08	L2	1	1,2,3
b)	Explain how the 'ability to pay' principle influences the design of a water tariff	08	L2	2	1,2
3. a)	Evaluate the reliability and cost-effectiveness of Decentralized Wastewater Treatment Systems compared to conventional centralized systems	08	L2	1	1,2

- b) Explain the mechanisms and performance of Constructed Wetlands in treating wastewater. Highlight key design considerations

08 L2 2 1,2,3

Unit – II

4. a) Evaluate the selection process for Tertiary Treatment when the goal is the highly reliable removal of pathogens for reuse purposes.

08 L2 3 1,2,3

- b) Explain the mechanisms and performance of granular-medium filtration. What role does filter media size play in ensuring its effectiveness?

08 L2 4 1,2,

5. a) Discuss the processes and challenges involved in the removal of toxic compounds and refractory organics from wastewater.

08 L2 3 1,2,3

- b) Explain the mechanisms and key design features of Micro Screening. Discuss its performance in removing fine suspended solids.

08 L2 4 1,2

6. a) Detail the construction and function of a Septic Tank and an associated Soak Pit. Evaluate this system's reliability in a rural context

08 L2 3 1,2,3

- b) Write a brief note on the removal of dissolved inorganic substances from wastewater, and what advanced treatment mechanisms are required

08 L2 4 1,2

Unit – III

7. a) Describe the technical and environmental considerations that guide the implementation of Wastewater Reuse and Reclamation projects

08 L2 5 1,2,3

- b) Explain the principles and application of Land Treatment Systems for wastewater reuse. Discuss the nutrient recycling aspect.

08 L2 5 1,2,3

8. a) Discuss the importance of Rural Water Supply and Sanitation, focusing on the core principles of Low-Cost Sanitation and Dry Sanitation methods

08 L2 5 1,2,3

- b) Detail the design and working principles of the following low-cost sanitation methods: Ventilated Improved Pit (VIP) Latrine and Aqua Privy

08 L2 5 1,2,3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
